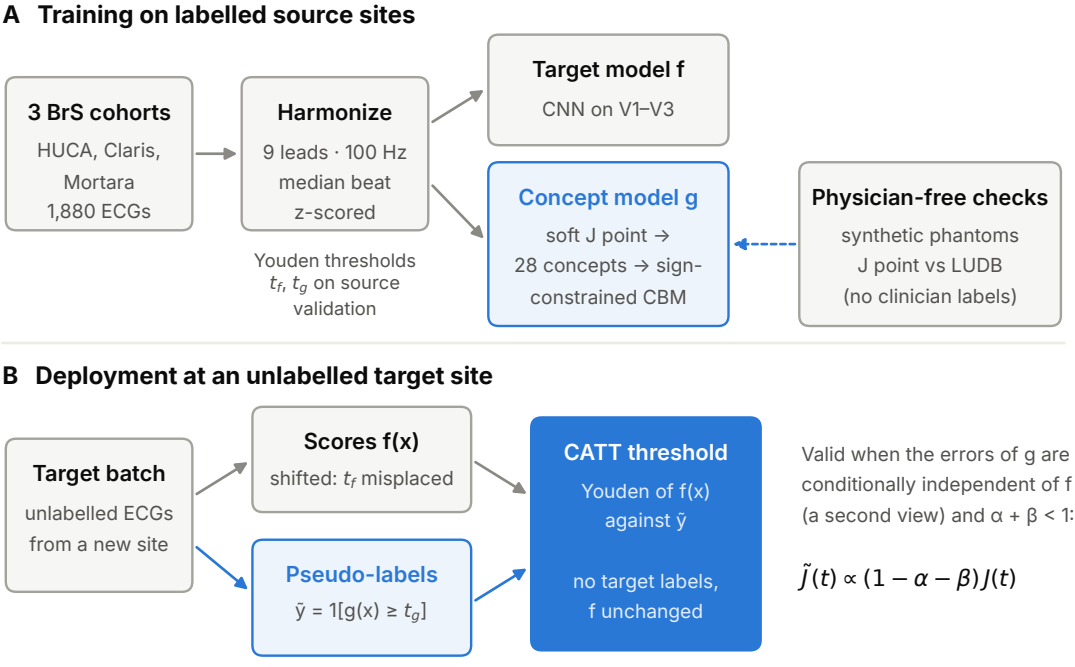
A Training on labelled source sites
3 BrS cohorts
HUCA, Claris, Mortara
1,880 ECGs
Harmonize
9 leads · 100 Hz
median beat
z-scored
Youden thresholds
t_f, t_g on source validation
Target model f
CNN on V1–V3
Concept model g
soft J point →
28 concepts → sign-constrained CBM
Physician-free checks
synthetic phantoms
J point vs LUDB
(no clinician labels)
B Deployment at an unlabelled target site
Target batch
unlabelled ECGs from a new site
Scores f(x)
shifted: t_f misplaced
Pseudo-labels
ỹ = 1[g(x) ≥ t_g]
CATT threshold
Youden of f(x) against ỹ
no target labels, f unchanged
Valid when the errors of g are conditionally independent of f (a second view) and α + β < 1:
$\tilde{J}(t) \propto (1-\alpha-\beta)J(t)$